As per ANSI standards, C++ places no limit on its length and therefore all the characters are significant.

Identifiers	Valid / Invalid	Reason for invalid
Num	Valid	
NUM	Valid	
_add	Valid	
total_sales	Valid	
tamilMark	Valid	
num-add	Invalid	Contains special character (-)
this	Invalid	This is one of the keyword. Keyword cannot be used as identifier names.
2myfile	Invalid	Name must start begins with an alphabet or an underscore

- You may use an underscore in variable names to separate different parts of the name (eg: ***total_sales*** is a valid identifier where as the variable called ***total sales*** is an invalid identifier).
- You may use capital style notation, such as ***tamilMark*** ie. capitalizing the first letter of the second word.

9.3.3 Literals (Constants)

Literals are data items whose values do not change during the execution of a program. Therefore Literals are called as Constants. C++ has several kinds of literals:

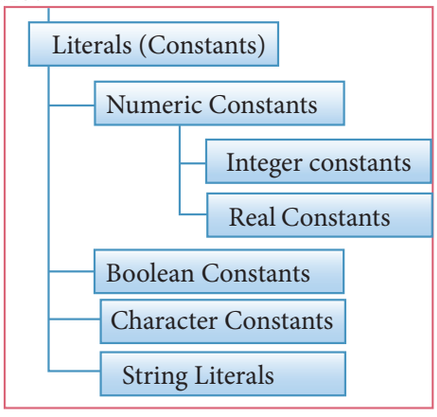


Figure 9.1 Types of Constants

Numeric Constants:

As the name indicates, the numeric constants are numeric values, which are used as constants. Numeric constants are further classified as:

1. Integer Constants (or) Fixed point constants.
2. Real constants (or) Floating point constants.

(1) Integer Constants (or) Fixed point constants

Integers are whole numbers without any fractions. An integer constant must have at least one digit without a decimal point. It may be signed or unsigned. Signed integers are considered as negative, commas and blank spaces are not allowed as part of it. In C++, there are three types of integer constants: (i) Decimal (ii) Octal (iii) Hexadecimal

(i) Decimal

Any sequence of one or more digits (0 …. 9)

Valid	Invalid
725	7,500 (Comma is not allowed)
-27	66 5(Blank space is not allowed)
4.56	9$ (Special Character not allowed)

If you assign **4.56** as an integer decimal constant, the compiler will accept only the integer portion of **4.56** ie. **4**. It will simply ignore **.56.**

Notes

If a Decimal constant declared with fractions, then the compiler will take only the integer part of the value and it will ignore its fractional part. This is called as "Implicit Conversion". It will be discussed later.

(ii) Octal

Any sequence of one or more octal values (0 …. 7) that begins with 0 is considered as an Octal constant.

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